HOW DAILY ACTIVITY SHAPES HEALTH

A data-driven approach using MedFit dataset

Goal: Understand how lifestyle impacts health

CONTEXT & PROBLEM

PEOPLE TRACK FITNESS BUT DON'T FULLY USE THE DATA

CAN WE EXTRACT MEANINGFUL HEALTH INSIGHTS FROM DAILY ACTIVITY?

OPPORTUNITY: TURN RAW DATA INTO ACTIONABLE INSIGHTS

DATASET DESCRIPTION

MedFit dataset (fitness & health tracking)

Variables: Steps, calories, heart rate, time

Structured time-based data for analysis

RESEARCH QUESTION

- How does daily physical activity influence health indicators?
- Focus: Calories burned and heart rate patterns
- Can activity levels indicate healthier lifestyle behavior?

HYPOTHESIS

Higher activity → higher calorie expenditure

Higher activity → increased heart rate

Consistent activity → healthier lifestyle patterns

PLANNED VISUALIZATIONS

1

Line chart → activity trends over time

2

Scatter plot → steps vs calories

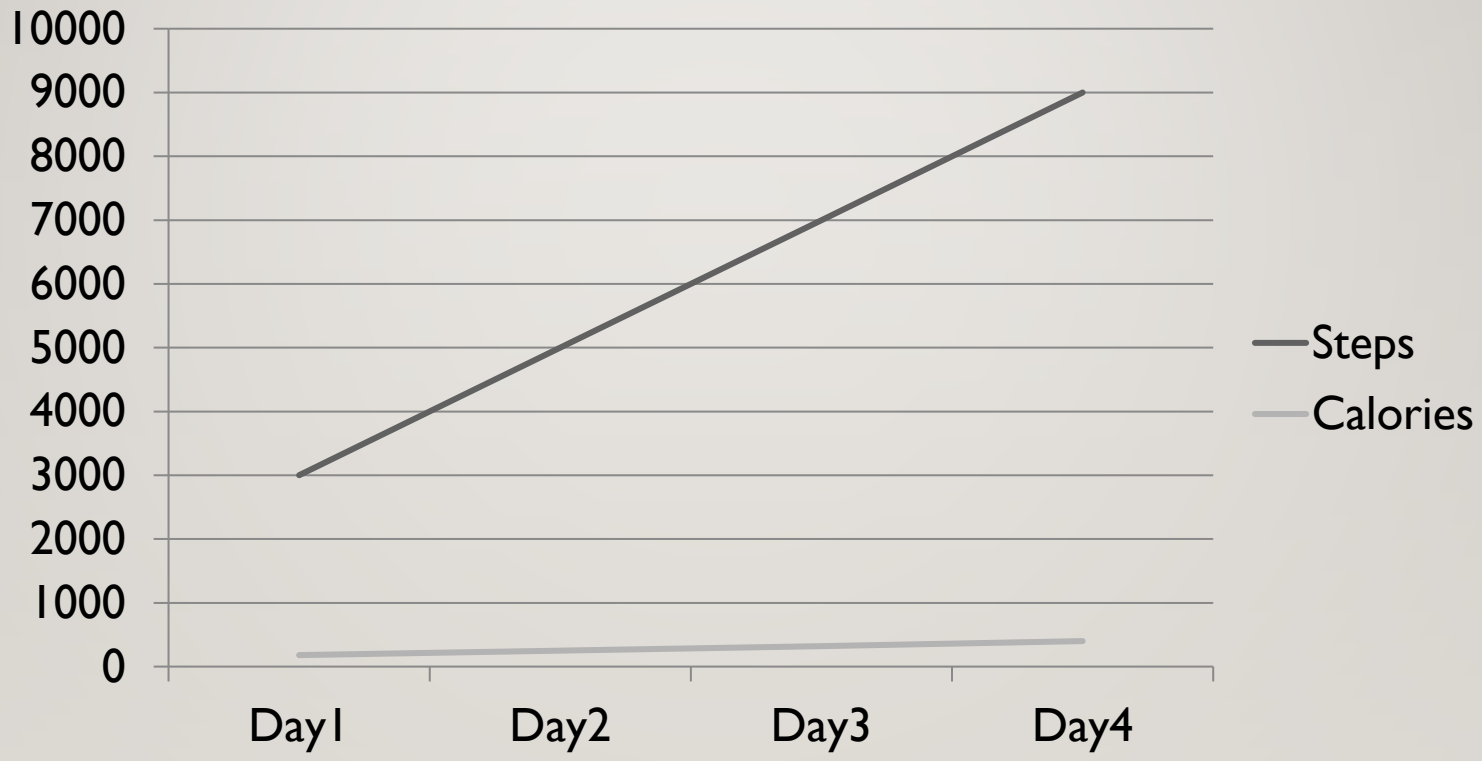
3

Bar chart → activity level comparisons

WHAT CAN WE DO WITH THIS DATASET?

- Analyze activity vs health relationships
- Identify healthy vs inactive behavior
- Track trends over time
- Support fitness app recommendations

SAMPLE VISUALIZATION: STEPS VS CALORIES



EXPECTED INSIGHTS

- Strong correlation between steps and calories
- Clear daily/weekly activity patterns
- Identification of active vs inactive lifestyles

BUSINESS / REAL-WORLD VALUE

- Fitness apps can personalize recommendations
- Users can optimize routines based on data
- Supports digital health and wellness solutions

METHOD APPROACH

- Data cleaning and filtering
- Descriptive statistics
- Visualization using tools like ggplot/Python

CONCLUSION

- Data reveals clear link between activity and health
- Simple insights can drive better lifestyle decisions
- Foundation for future health-focused applications